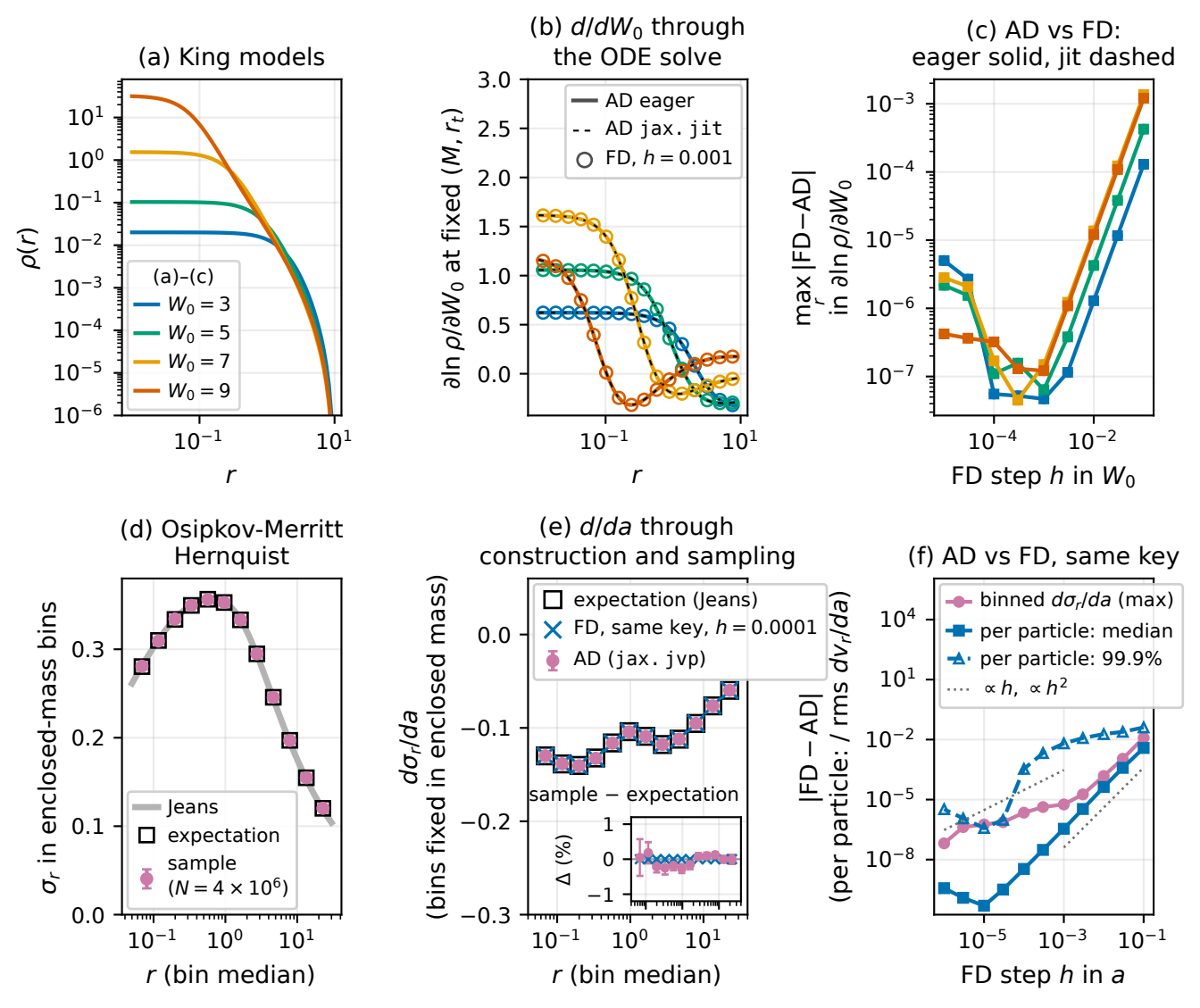
(a) King models
$\rho(r)$
(a)–(c)
$W_0 = 3$
$W_0 = 5$
$W_0 = 7$
$W_0 = 9$
r
(b) d/dW_0 through the ODE solve
AD eager
AD jax.jit
FD, $h = 0.001$
$\partial \ln \rho / \partial W_0$ at fixed (M, r_t)
r
(c) AD vs FD: eager solid, jit dashed
$\max_r |\mathrm{FD} - \mathrm{AD}|$ in $\partial \ln \rho / \partial W_0$
FD step h in W_0
(d) Osipkov-Merritt Hernquist
σ_r in enclosed-mass bins
Jeans
expectation
sample ($N = 4 \times 10^6$)
r (bin median)
(e) d/da through construction and sampling
expectation (Jeans)
FD, same key, $h = 0.0001$
AD (jax.jvp)
$d\sigma_r/da$ (bins fixed in enclosed mass)
sample − expectation
Δ (%)
r (bin median)
(f) AD vs FD, same key
binned $d\sigma_r/da$ (max)
per particle: median
per particle: 99.9%
$\propto h$, $\propto h^2$
$|\mathrm{FD} - \mathrm{AD}|$ (per particle: / rms dv_r/da)
FD step h in a